

CASA0015: Mobile Systems & Interactions

Assessment Guidelines

You are required, as an individual, to develop and produce a mobile application which showcases the skills you have acquired on this module. Alongside this mobile app you will also submit:

- The source code for the mobile app submitted on GitHub.
- A video or gif showcasing the user interface and screens of the application within the GitHub repository.
- The GitHub repository should show all commits and show an ongoing and iterative progress through the development process.

The mobile app should contain multiple views and allow the user to perform some function or interaction with a physical device or onboard sensors and be focused on a theme within the Connected Environment. The application must be written in Flutter and be able to be deployed on multiple devices (iOS or Android). The mobile application should be tested, and compile, on native hardware and include some sort of user testing scenario as well as make use of external cloud services to extended the functionality of the app. Marks will be awarded for good design practices and creative uses of the mobile device interaction to perform a specific function in response to the brief.

Assessment Weighting

This coursework represents 100% of the overall module assessment.

Stages of Assessment

Mobile Application - Interactive mobile application for iOS or Android - **80%**

Final Presentation - **20%**

Key Dates for 2021/2022

Pitches of project idea – **See Moodle**

Mobile Application submission – **See Moodle**

Presentation of Final App – **See Moodle**

Assessment Outline

Pitches – unassessed but required

5 min presentation of app idea with 3 minutes for questions.

This presentation is a rough overview of your idea for your mobiles, with wireframes and storyboarding of the user experience, the identification of the functionality of the mobile app, and what problem the app is trying to solve. It is important to note that students will not be held to what they present, and amendments to the project can be made after this crit. The aim of this informal presentation is to ensure sure students have an achievable goal within the time frame and to source valuable feedback from tutors and the rest of the class.

This pitch is **not assessed but is a required element of the coursework.**

Mobile Application – 80%

2000 equiv. word count

This mobile application is the main component of the assessed individual work for this module. Students are required to create a mobile app within the theme of Connected Environments, with multiple views that convey a narrative and solves a problem for the user. As a bare minimum, the mobile app should contain a introductory splash screen, use some of the on-board sensors to monitor or detect the physical environment and log that data over time. The choice of application or problem solved is up to the student and is open-ended, however the app should solve a problem within the Connected Environments and be broadly within the theme of the overall degree. The problem statement that the app is solving should be outline and highlighted within the students GitHub repository.

The mobile application should be interactive, respond to user input and pull or accept push data from various sources such as an online repository of data (through an API, MQTT or related technology). The application can solve any problem the student wishes but marks will be given for creativity and identifying a mobile pattern which aids it users and triggers repeat interactions at different times. You are expected to demonstrate the full extent of knowledge of Flutter and other libraries you have been taught during this module. You are furthermore expected to make use of those tools taught on the course in completing the mobile application work (e.g. Flutter, Sketch, iOS/Android).

Students are encouraged to use any frameworks or libraries (available through the Flutter Widget catalog etc.) to improve the functionality of the app. Students are reminded that the course is assessed on the skills and techniques learnt during the course, as such, we therefore expect all the source code to be Dart.

The source of the mobile app and supporting backend service should be submitted through a GitHub repository showing the commit history of the application which will be checked before submission. Your GitHub Repository should be tagged as a Flutter repository and also have screenshots of your app included in the README.md page along with a meaningful explanation of what your app does.

Important - You must show daily or weekly progress through the commit history for full credit for this module. Students are also reminded that they should not download, purchase or use a commercial mobile app as they are expected to write the Dart, Flutter Application within as part of the assessment of this module.

The mobile application will be assessed by the following criteria:

- User Interface and Experience - **20%**
 - Use of compelling and appropriate or interactive elements / widgets - **30%**
 - Exploratory, 'storytelling' nature of application - **20%**
 - Use of API or Service to enhance the Application - **15%**
 - Functionality of the application to solve a problem - **15%**
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Presentation of Final App – 20%

15 minute presentation with 5 minutes for Questions

Each student must present their completed mobile application to the rest of the class that discusses the design from inception to development, and the execution of the project.

The presentation should include a video or gif of the mobile application (max 3 minutes in length), either from a physical device or simulator, and a landing page, written in HTML which advertises the application. The landing page can be submitted on GitHub pages with the source code of the mobile app and use a theme available from many online, open source, free landing page themes. The landing page should showcase the functionality of app and how the app solves a problem for the user. The presentation should also include the collection and handling of any data, user experience journey through storyboarding and wireframes design (UX), development of the app, and interactivity with external APIs or services used within the project. A conclusion should detail how, if given more time, the student would improve the mobile application.

A breakdown of how the presentation will be marked is as follows:

- Video or gif Showcasing the Application - 3 mins max - **20%**
- Landing Page of the Application - **20%**
- Overview of Mobile App - **50%**
 - Design, storyboarding, outline of application
 - Data collection, handling and management of user data
 - Interactivity of application
 - Technical integration between mobile app and services
- Style of the presentation - **10%**

If you have any major issues with your assessment you should contact the module leader Steven Gray - steven.gray@ucl.ac.uk.

Design Inspiration

Examples of Good Mobile App Design

Here are a few examples to get your minds working and in the right mindset. We do not expect the level of quality or detail that some of these projects go into. However, this list should serve as a guide of the types of mobile apps we are looking for you to create:

- Flutter Showcase - <https://flutter.dev/showcase>
- Flutter Open Source Apps - <https://github.com/tortuvshin/open-source-flutter-apps>

Flutter Resources

Here are a few Flutter resources we expect students to take advantage of to help create their mobile app:

- Flutter - <https://flutter.dev>
- Flutter Docs - <https://flutter.dev/docs>
- Awesome Flutter - <https://github.com/Solido/awesome-flutter>
- Pub.dev - Dart Packages - <https://pub.dev>
- DartPad - <https://dartpad.dartlang.org>
- Dart by Example - <https://www.jpryan.me/dartbyexample/>

Firebase Resources

- Firebase List on Github - <https://github.com/vsergiu/firebase-resources>
 - Awesome Firebase - <https://github.com/jthegedus/awesome-firebase>
 - FlutterFire - <https://github.com/FirebaseExtended/flutterfire>
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